

US-1

As a CEDARS Accommodation Specialist, I want to manage accommodation details, so that accommodation can be manipulated and changed in UniHaven.

Sprintable stories:

1. As a CEDARS Accommodation Specialist, I want to get a database, so that the accommodation details can be stored and viewed.
 - i. Design database schema for Accommodation model with all required fields (ID, title, description, type, current status, detailed address, price, number of beds, number of bedrooms, owner (foreign key), location (foreign key))
 - ii. Design database schema for Owner model with all required fields (ID, Name, contact method, contact information)
 - iii. Design database schema for Location model with all required fields (ID, Building name, building location, latitude, longitude)
 - iv. Build an API endpoint for creating listing (POST /admin/accommodation/<model>/1/change/), retrieve listing (Get /static/admin/...) and remove listing (POST /admin/accommodation/<model>/1/delete/...)
2. As a CEDARS Accommodation Specialist, I want to upload accommodation details, so that new accommodation can be shown in UniHaven.
 - i. Perform several forms for (accommodation, owner and location) model for inputting record for uploading to database
 - ii. Performing the submission of data to the database.
3. As a CEDARS Accommodation Specialist, I want to delete accommodation details, so that unserved accommodation can be removed in UniHaven.
 - i. Perform removal of record for accommodation, owner and location models
4. As a CEDARS Accommodation Specialist, I want to view accommodation details, so that I can check for maintenance of data.
 - i. Perform checking records for accommodation, owner, location models

US-2

As a CEDARS Accommodation Specialist, I want to manage reservations: make reservations, cancel reservations, and view detailed information about selected accommodations, so that I can plan or cancel contract signing in time.

Sprintable stories:

1. As a CEDARS Accommodation Specialist, I want to get a database, so that the accommodation details can be stored and viewed.
 - i. Design database schema for reservation models with all required fields (ID, title, description, type, current status, price, number of beds, number of bedrooms, price, address, member(foreign key), accommodation(foreign key))
 - ii. Design database schema for Member model with all required fields (ID, Name, contact method, contact information)
 - iii. Build an API endpoint for creating listing (POST /admin/accommodation/<model>/1/change/), retrieve listing (Get /static/admin/...) and remove listing (POST /admin/accommodation/<model>/1/delete/...)
2. As a CEDARS Accommodation Specialist, I want to make reservations, so that I can secure it for a student's stay.
 - i. Create serializers for reservation data. Define the structure for reservation data to be serialized. Implement serializers that include all relevant fields (e.g., user ID, accommodation ID, dates).
 - ii. Design reservation model. Create a reservation model that includes proper relationships to Accommodation and User. Ensure that the model captures all necessary attributes.
 - iii. Build API endpoint. Implement the API endpoint for creating reservations (POST/API/reservations/{ID}). Ensure that the endpoint accepts the necessary data and returns appropriate responses. Implement the API endpoint for viewing single accommodation (GET/API/accommodations). Update the endpoint to include reservation availability status if necessary.
3. As a CEDARS Accommodation Specialist, I want to cancel reservations at any time points during the contract period, so that students have a way out if they regret their decision after signing the contract.
 - i. Perform removal of record for reservation, member models
 - ii. Build API endpoint. Implement the API endpoint to retrieve a list of reservations for the authenticated user (GET/API/reservations/{ID}). Implement the API endpoint for cancelling a reservation (DELETE /API/reservations/{ID}). Ensure the endpoint handles successful cancellations and returns appropriate responses.
4. As a CEDARS Accommodation Specialist, I want to view detailed information about selected accommodations, so that I can access it for my needs.
 - i. Access data from database developed in US-1 according to the selected accommodations.

Ensure the database schema supports all necessary accommodation fields. Implement data retrieval logic to access details of selected accommodations.

- ii. Implement status tracking for accommodations (Pending, Confirmed, Cancelled). Define status categories for accommodations (Pending, Confirmed, Cancelled). Update the database schema to include a status field. Implement logic to update accommodation statuses based on reservations.
- iv. Design detailed view serializer Define a serializer for accommodations that includes all relevant fields (e.g., images, amenities, pricing, status). Implement the serializer in the backend.
- v. Build API endpoint. Create the API endpoint to fetch details of a single accommodation (GET/API/accommodations/{ID}). Ensure the endpoint returns the accommodation data in the correct format.

US-3

As an HKU member, I want to search for accommodation based on various criteria (accommodation type, period of availability, number of beds, number of bedrooms, price, distance from a selected HKU campus or premises), so that I can view details of accommodation and find suitable options.

Sprintable stories:

1. As an HKU member, I want to search for accommodations matching certain requirements (type, availability, number of beds, number of bedrooms, price).
 - i. Design filtering parameter schema:
 - accommodation type: apartment, house, single room, share room
 - period of availability: available, unavailable
 - number of beds: a list of 1, 2, 3, 4 or above
 - number of bedrooms: a dropdown list of 1, 2, 3 or above
 - price: \$3000 or below, \$3000-\$5000, \$5000-\$7000, more than \$7000
 - ii. Implement filtering logic for each criterion:
 - accommodation type: matching the criteria with the database established in US-1
 - period of availability: matching the period with the database established in US-1, filtering those whose available period covers the user-input period.
 - number of beds: matching the number of beds with the database established in US-1
 - number of bedrooms: matching the number of bedrooms with the database established in US-1
 - price: matching the price with the database established in US-1, filtering those whose price is within the user-input price range
 - iii. Build API endpoint for searching (GET/API/accommodations/search)
2. As an HKU member, I want to view the list of accommodation matching my requirements after searching.
 - i. Create serializer for search results